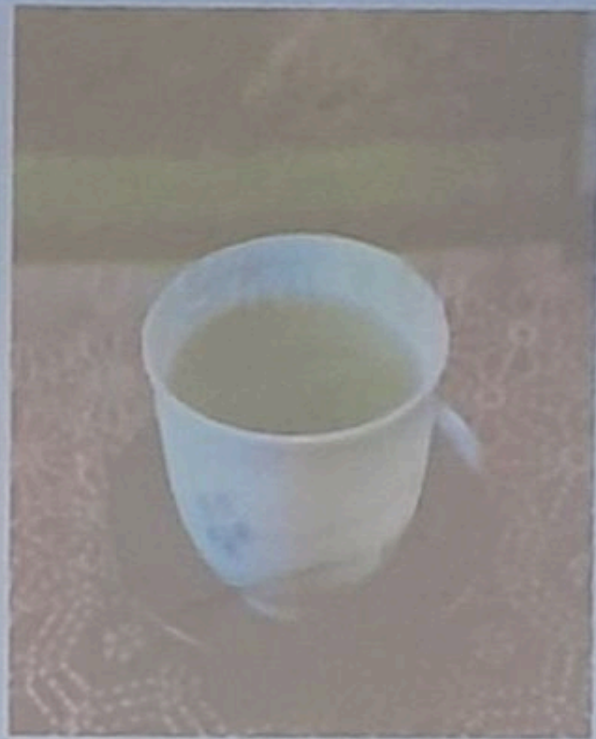
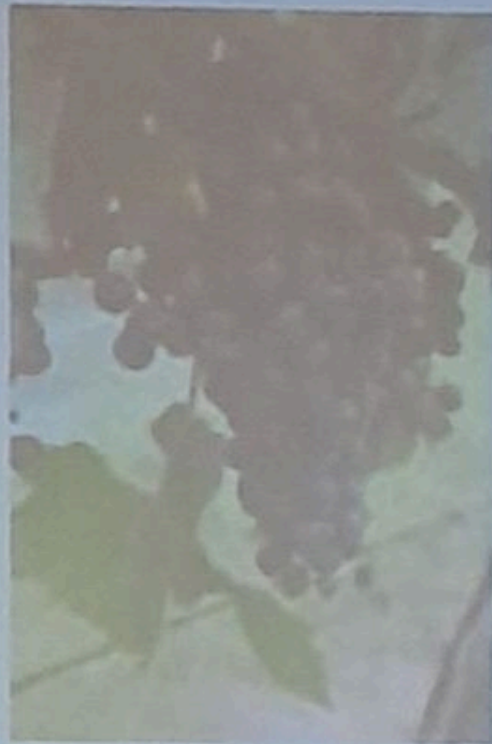


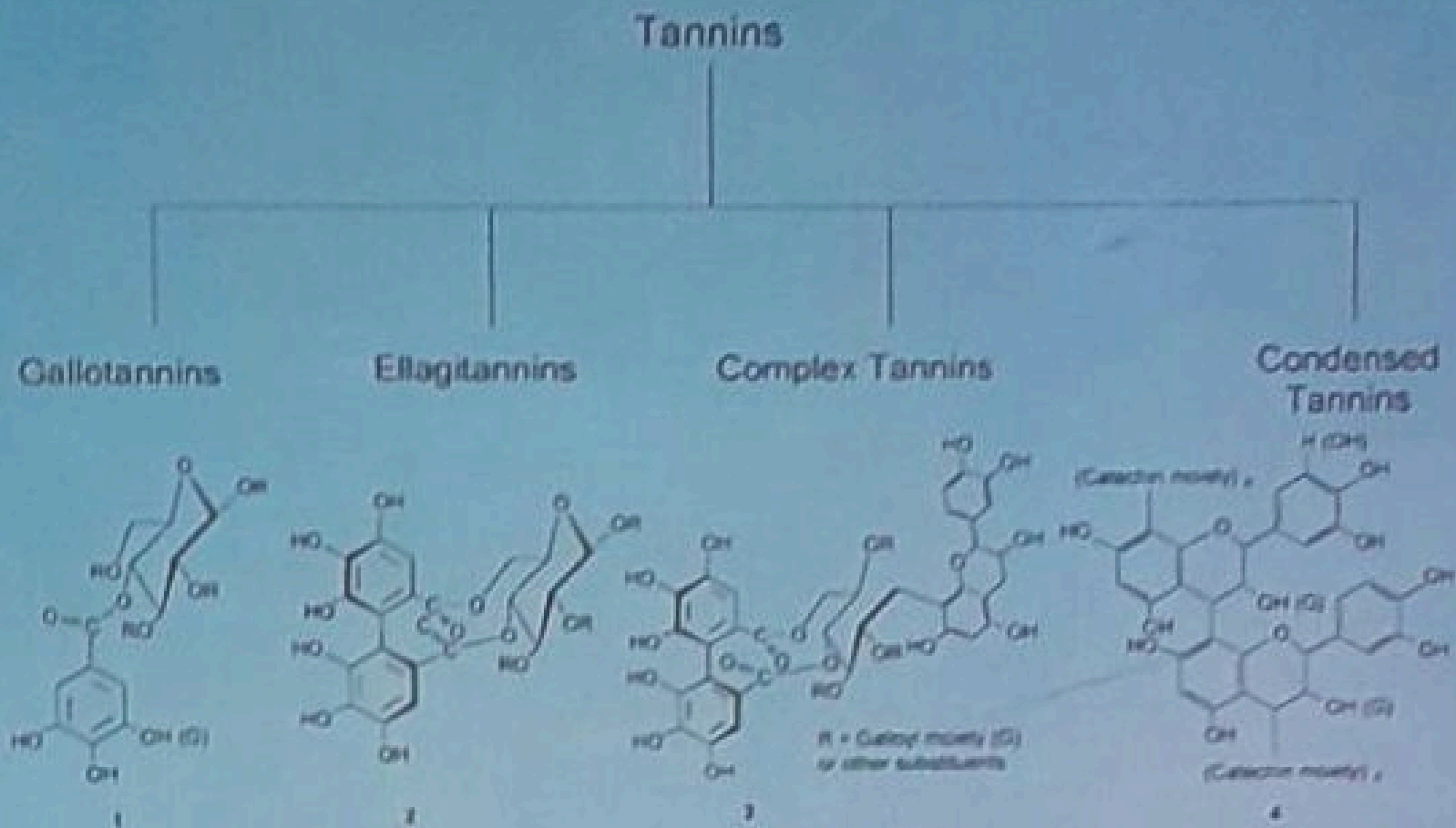
Chapter 4: TANNINS



- The term 'tannin' was first applied by Seguin in 1796
 - To combine with protein of animal hides, prevent their putrefaction and convert them into leather.
- Are complex, non-nitrogenous derivatives of polyhydroxy benzoic acids
- molecular weight is usually high
- produce greenish black or blue color with ferric chloride
- precipitates with solutions of cellulose, pectins, gelatin, proteins, alkaloids and salts of heavy metals

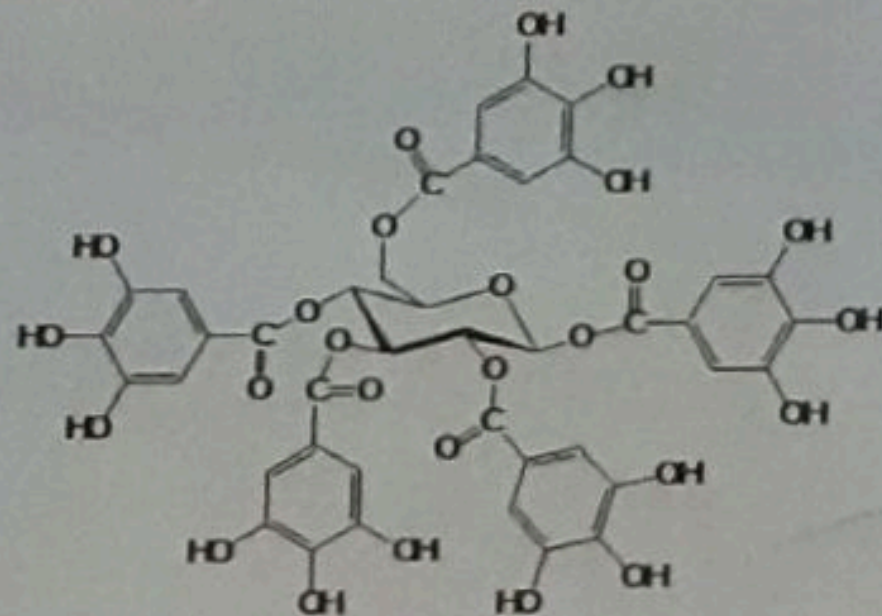
- are **non-crystalizable** compounds prevent others from crystallization
- are acidic
- are soluble in water, alcohols, hydroalcoholic solutions, acetone and alkalis
- insoluble in organic solvents such as ether, benzene etc.
- cause agglutination of red blood cells
- are mainly found in the cell vacuoles or in the cuticle (here cells don't interfere with plant processes:
 - only after death and cell breakdown do they become active in metabolic processes)

Classification of Tannins



A. Hydrolysable Tannins

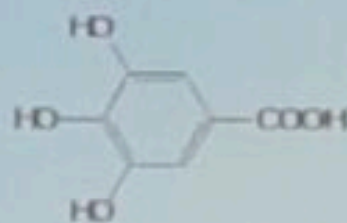
- are hydrolyzed by weak acids or weak bases/enzymes (tannase) to produce carbohydrate and phenolic acids



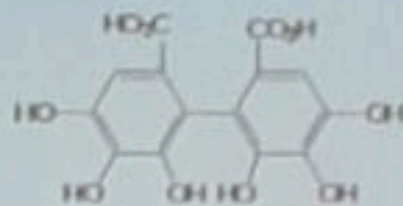
penta-O-galloyl-β-D-glucose

Hydrolysable Tannins Cont...

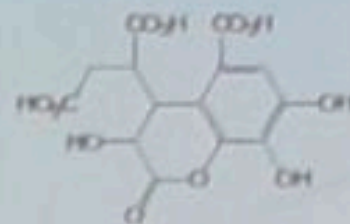
- Based on the nature of the phenolic acid, hydrolysable tannins are further classified into:
 - gallitannins (gallic acid)
 - ellagitannins (hexahydroxydiphenic or chebulic acid)



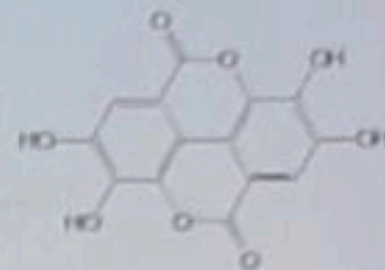
gallic acid



hexahydroxydiphenic acid



chebulic acid

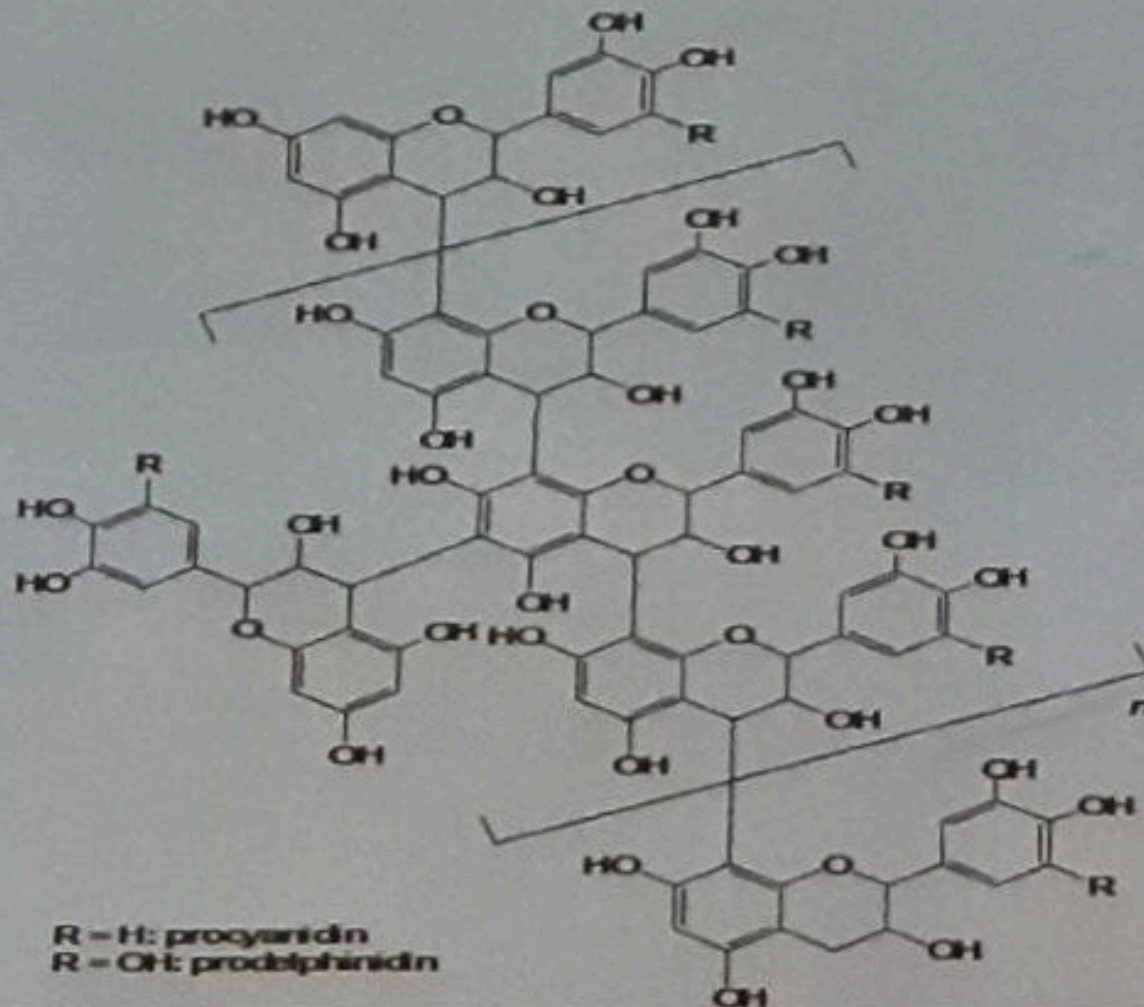


ellagic acid

B. Condensed Tannins (proanthocyanidins tannins)

- are polymeric flavans
- they tend to polymerize in strong acids or oxidizing agents producing insoluble red or brown colored products known as phlobaphenes
- most are formed by the condensation of flavan-3-ols (such as catechin) or flavan-3,4-diols (such as leucocyanidin)

Condensed tannins

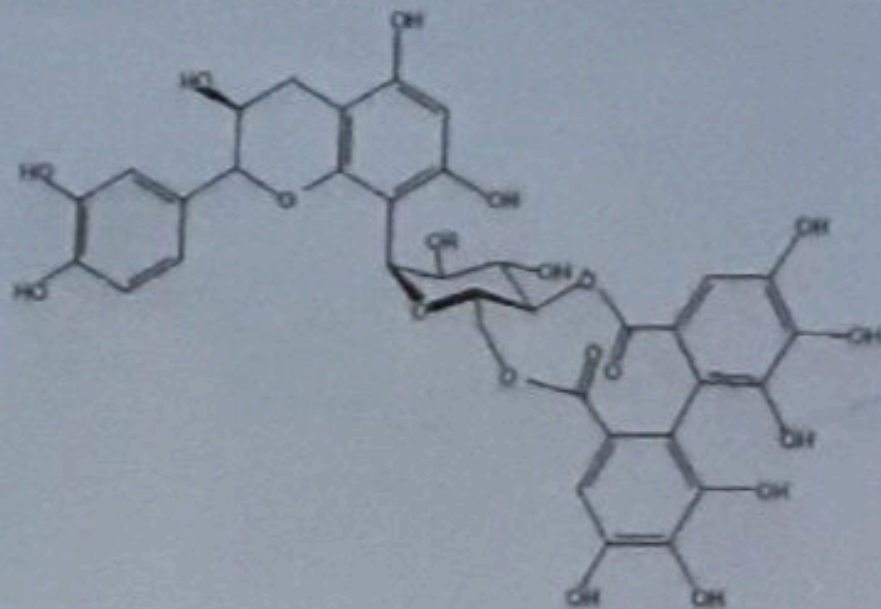


Comparison

	Hydrolysable (Pyrogallol)	Condensed (Catechol)
Hydrolysis	Hydrolysable	Non-Hydrolysable
Dry distillation	Gives pyrogallol	Gives catechol
FeCl₃	Blue-black colour	Green colour
Br₂ water	No ppt.	Ppt
Examples	Gall, Clove, Rhubarb	Cinnamon, Tea, Henna.

C. Complex Tannins

- are modified ellagitannins resulting from the addition of a flavonoidal moiety



D. Pseudotannins

- are compounds of low molecular weight
- give some of the reactions of tannins
- usually found in combination with tannins

Examples:

- catechins
- chlorogenic acid
- ipecacuanic acid

Uses of Tannins

- **Medicinal and pharmaceutical:**

- as astringent (in superficial wounds)
 - they enhance tissue regeneration
- as antidotes in poisoning of heavy metals, alkaloids and some glycosides
 - for the protection of inflamed surfaces of mouth and throat
 - as antidiarrheal
 - also exhibit antibacterial, antifungal and antitumor activity

- **Commercial:**

- are widely used in leather industry (tanneries)
- to clarify wine and beer, reduce viscosity of oil-well drilling mud

Chapter 5:

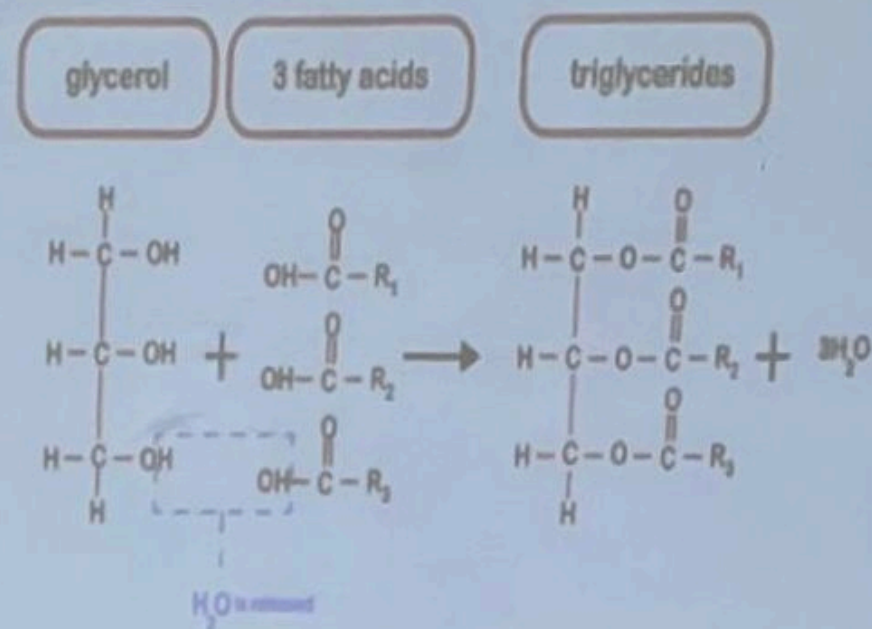
Lipids and waxes

- The term **Lipids or fatty esters.**

- Are esters of **alcohol (trihydride)** and an **acid (saturated or unsaturated)**
 - Includes fixed oils, fats and waxes (**simple lipids**),
 - And also phosphatides and lecithins (**complex lipids**), which contain phosphorus and nitrogen in addition
-
- Widely distributed in both the plant and animal kingdoms
 - **fruits and seeds**

Fixed Oils and Fats

- Fixed oils and fats are heterogeneous mixtures of different esters of **glycerol** with higher **long chain fatty acids** (4-24 carbon atoms)
- Fixed oils and fats differ in the proportion of saturated fatty acids occurring in their triglycerides
 - Fixed oils** have higher proportion of **unsaturated fatty acids** in their triglycerides
 - Fats** have higher proportion of **saturated fatty acids** in their triglycerides
 - The **complete hydrolysis** fixed oils and fats yield **glycerol** and **mixtures of fatty acids**



Physical Properties of Fixed Oils and Fats

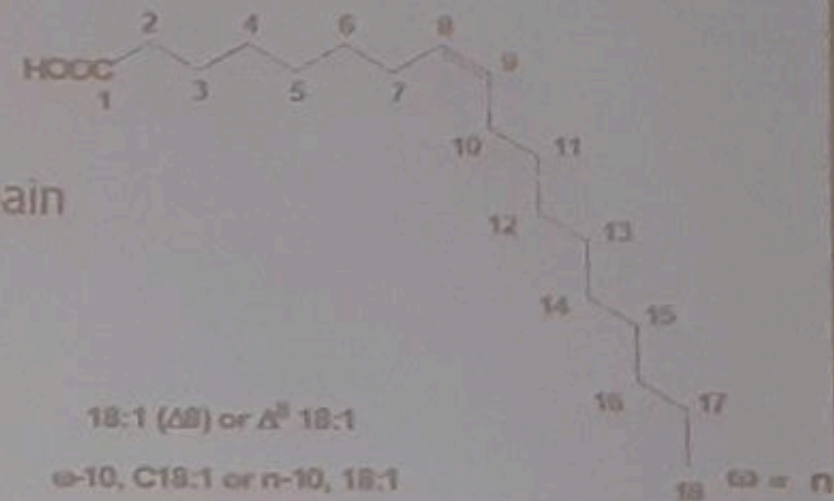
- At room temperature, **fixed oils are liquids** and **fats are solid or semisolid**
- Greasy substances
- have **lower specific gravity** than water
- Insoluble in **water and alcohol**
- Soluble in **non-polar solvents** such as petroleum ether, ether, hexane, chloroform, benzene etc.
- **not volatile**
- Melting points and boiling points are not usually sharp.
- Pure fats and oils are **colorless and odorless**
- Color and odor is always due to plant pigments or contaminants

Fatty Acids Found in Fixed Oils and Fats

- Responsible for **pharmacological** activity
- In fixed oils and fats, fatty acids **do not normally occur free** in significant amount
- They are obtained by hydrolyzing the triglycerides of fixed oils or fats
- They have a **single carboxyl group** (Monobasic acids) and a long non-polar hydrocarbon tail
- Naturally occurring fatty acids differ from one another
 - by their **chain length**
 - in the **presence, number and position** of their double bonds
 - in the presence or absence of **alkyl branches**

- Nearly all fatty acids in nature have an even number of carbon atoms
 - According to chain length FAs can be
 - short chain FA: **2-4 carbon atoms**
 - medium chain FA: **6 –10 carbon atoms**
 - long chain FA: **12 – 24 carbon atoms**
- Fatty acids with 16 and 18 carbons are the most abundant

- In nature, **unsaturated fatty acids** are twice as abundant as saturated fatty acids
- Double bond is always **cis** in natural fatty acids
 - This lowers the melting point due to "kink" in the chain
- **non-conjugated double bonds**
- Most polyunsaturated fatty acids are **ω -6 fatty acids**
- **ω -3 fatty acids** are found mainly in fish and fish products. They also occur in linseed oil
- Longer chain and saturation increases melting point of fatty acids

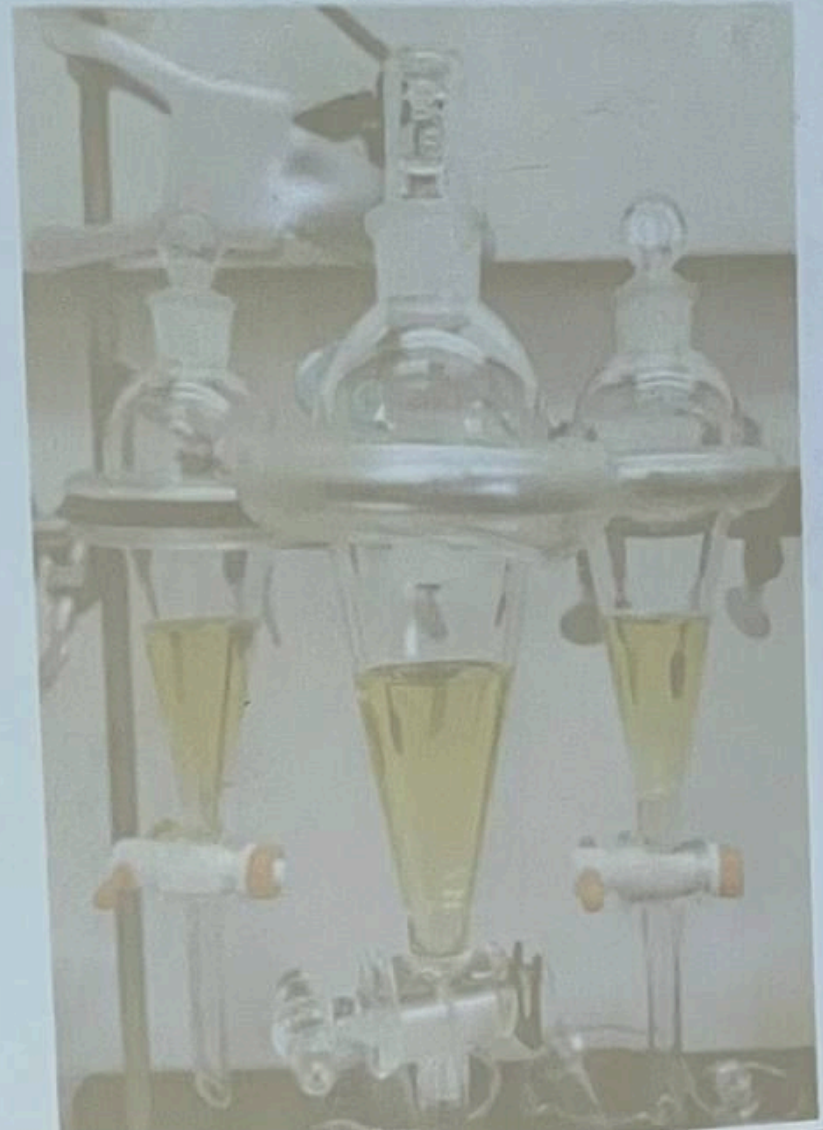


Extraction

From Plants

- **Expression method:** In expression method the seeds of the oil-bearing plant are crushed between two heavy rollers to squeeze the oil out
- If the heavy rollers are heated the method is called **hot expression**
 - Hot expression increases yield but low quality
- If the heavy rollers are not heated the method is called **cold expression**
 - Cold expression is usually associated with low yield but gives **good quality** oil with low level of contamination

- **Solvent extraction:** could be used for seeds
 - It gives good yield but poor quality products at the same time



From Animals

- By heating the animal tissue with steam, the fat is melted and separated from the remaining matter.
- Alternatively adipose tissues are cut and mixed with hot water to melt the pure fat which is then separated by decanting.

Quantitative tests

- A number of quantitative tests are commonly used to evaluate fixed oils and fats
 - **Acid value:** refers to the number of mg of **KOH** required to **neutralize the free acids** in 1 g of the oil; high acid values arise in rancidified oils
 - **Saponification value:** the number of mg of **KOH** required to **neutralize the free acids** and hydrolyze the esters in, 1 g of the oil
 - **Ester value:** is the difference between the saponification and acid values.
 - **Iodine value:** gives a measure of the unsaturation of the oil.

Refining Process for Fixed Oils and Fats

Degumming:

- This is the first step in the refining process of many oils
- It is carried out to remove **phosphatides** (**phospholipids**)
- The oil is mixed with water to hydrate the phosphatides, which are then removed by centrifugation
- Phosphoric acid, citric acid or silica gel are added to enhance the process

Alkali refining:

- The degummed oil is then treated with alkali to remove free fatty acids
 - Which lead to rancidity and oxidation
- The free fatty acids will form a soap that can be removed by centrifugation
- Any residual soap is removed by hot water washing

Bleaching:

- Trace metals, pigments (e.g. chlorophyll) and oxidation products are removed using bleaching clays, which adsorb them
- Bleached oils are nearly colorless

Winterization:

- Ordinary triglycerides of myristic acid, palmitic acid and stearic acid solidify at low temperatures, which results in cloudy appearance
- Some oils are thus chilled at 0°C with gentle agitation to precipitate high melting fraction
- The fraction which settles out is called stearin

Deodorization:

- Deodorization is a steam distillation process carried out in vacuum to remove volatile compounds from the oil

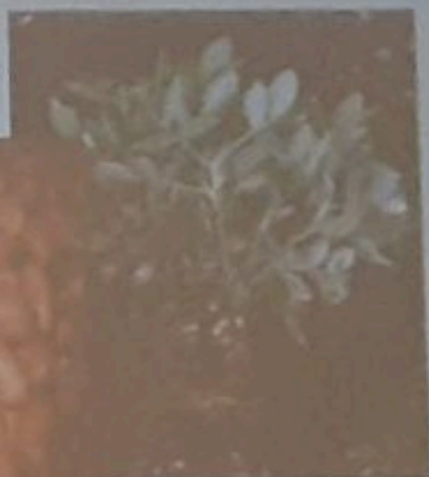
Commercial Applications of Fixed Oils and Fat

- As food
- Lubricants
- Soap production
- In paint and varnish manufacturing (drying oils)

Examples of Fixed Oils and Fats Used in Pharmacy and Medicine

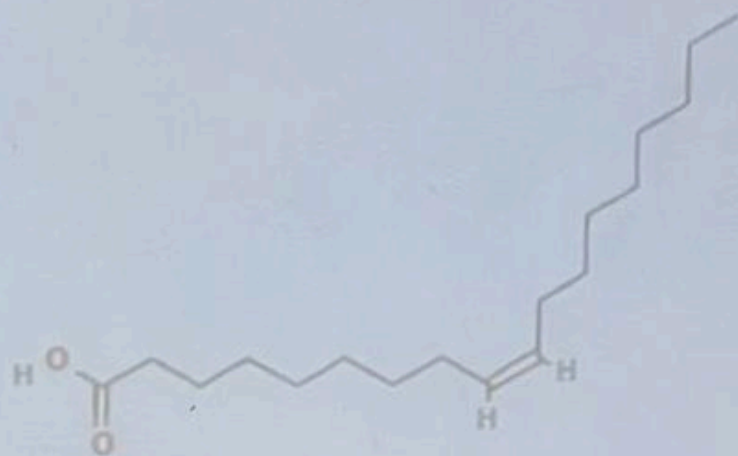
Peanut oil (arachis oil)

- the seed kernel of the cultivated varieties of *Arachis hypogaea* (Fabaceae)
- cold extraction followed by solvent extraction of the cake
 - two oil extracts are mixed before purification
- colorless or a pale-yellowish liquid



Fatty Acid Composition

- 8-20% palmitic acid (16:0)
- 1-4% stearic acid (18:0)
- **56-86% oleic acid (18:1)**
- 4-20% linoleic acid (18:2)
- Minor fatty acids are:
 - arachidic acid (20:0),
 - behenic acid (22:0)
 - lignoceric acid (24:0)

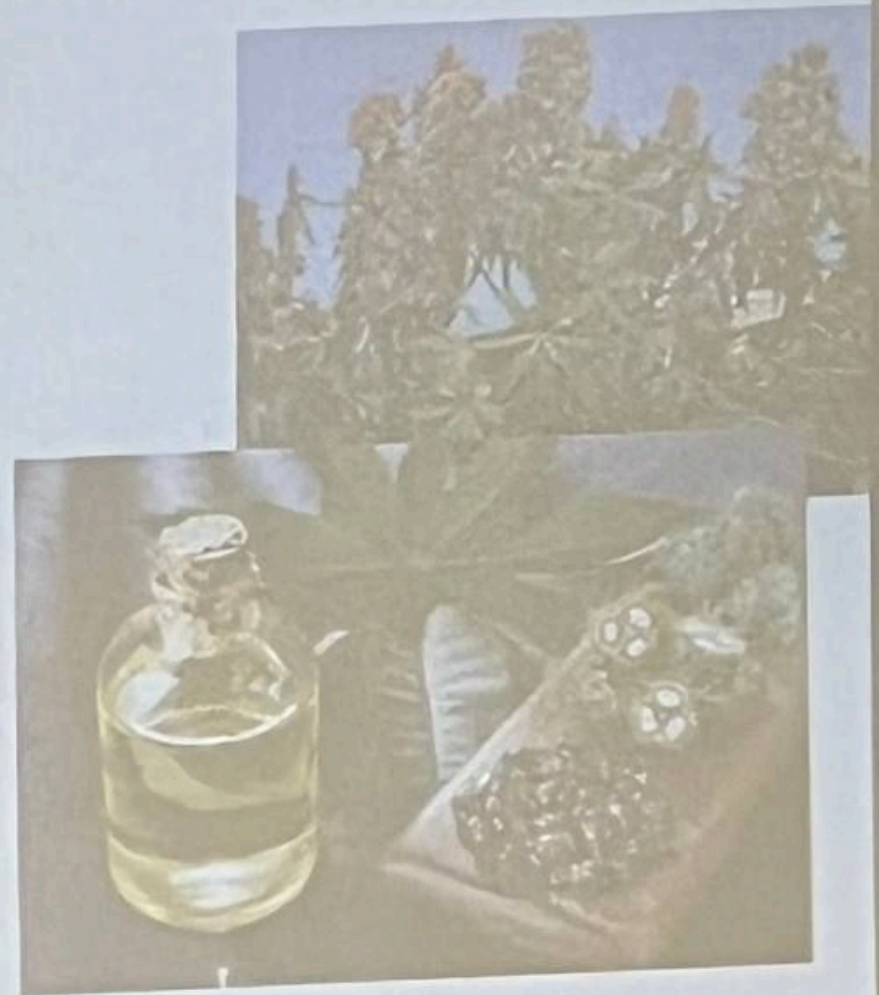


Use

- a vehicle for intramuscular injections
- preparation of margarine
- Note:
 - Because of its similarity with olive oil, peanut oil is commonly used to adulterate olive oil
- may cause serious allergic reaction

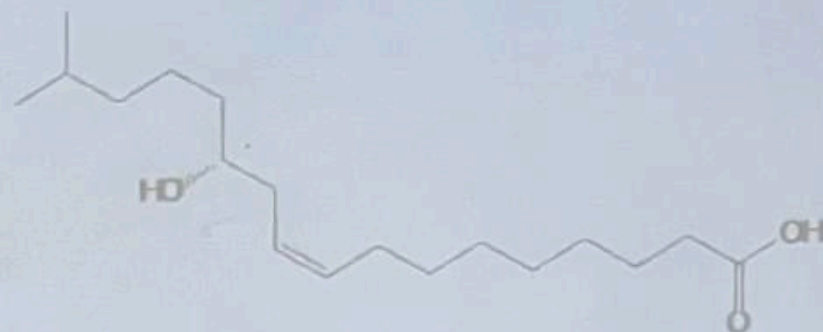
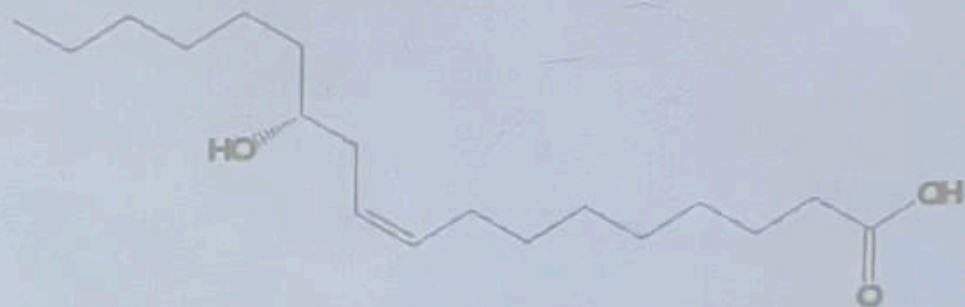
Castor Oil

- cold expression from the ripe seeds of *Ricinus communis* (Euphorbiaceae)
- After extraction, castor oil has to be steamed, filtered and bleached
 - Steaming is required, in order to **denature ricin**
 - an extremely toxic **glycoprotein (lectin)**
- Medicinal castor oil is colorless or pale yellow liquid of extremely high viscosity



Fatty Acid Composition

- **90% ricinoleic acid**
- 1% isoricinoleic acid
- (3%) oleic acid
- (3-4%) linoleic acid, dihydroxystearic acid, palmitic acid and stearic acid



Uses

- The oil is **stimulant purgatives**
 - This is due to **ricinoleic and isoricinoleic acids** which are liberated from their triacylglycerol forms by **pancreatic lipase** in the duodenum
 - Act on the walls of the small intestine
 - onset of action is thus delayed by **1-2 hrs**
 - It is applied in food poisoning and before intestinal examination

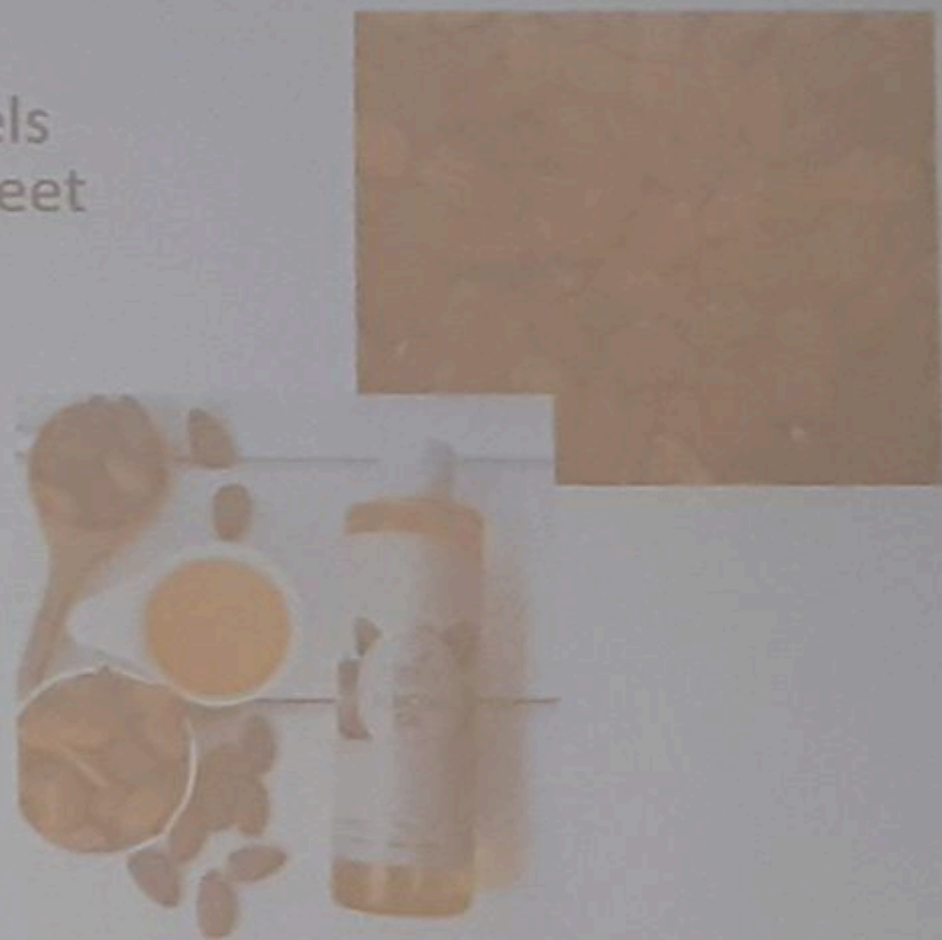
- Externally as emollient.
- Because of its high viscosity, it is used as stiffening agent in ointments and creams.
- Polyethoxylated castor oil is used in some intravenous preparations, which contain drugs of low water solubility.
- Also used for the preparation of lubricants

Remark

- Note that the cake obtained after the extraction of oil is not fit for use as cattle food
 - because it contains high amount of ricins
 - the cake could be used as chicken food
- Children should not be allowed to play with castor beans

Almond Oil

- cold expression from the seed kernels of *Prunus amygdalus* var. *dulcis* (sweet almond) or *Prunus amygdalus* var. *amara* (fam: Rosaceae).
- Almond oil is pale yellow liquid
- Fatty acid composition
 - 4-9% palmitic acid (16:0)
 - 1% stearic acid (18:0)
 - 62-86% oleic acid (18:1)
 - 20-30% linoleic acid (18:2)



Use

- In **cosmetology** and **dermatology**
 - Relieve skin irritation and sensation of heat
- Could also be used as a vehicle for oily injections

Remark:

- HCN may occur in bitter almond oil and therefore its presence and level should be determined

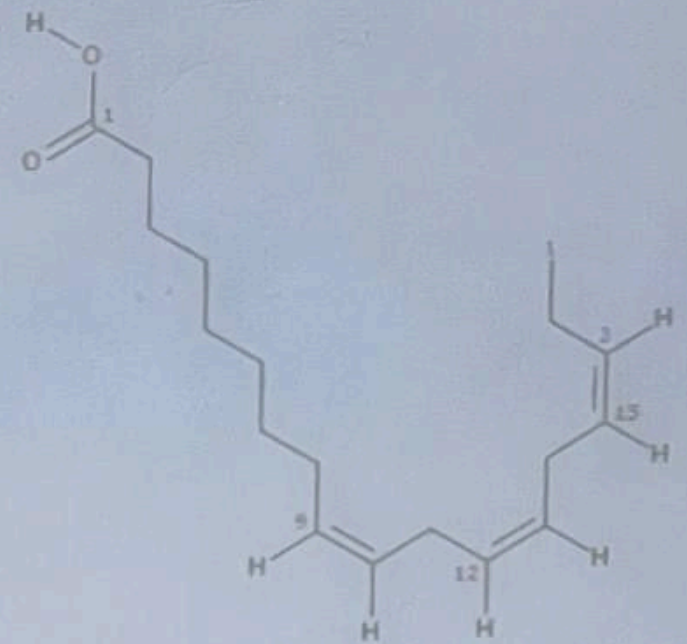
Linseed Oil (Flaxseed Oil)

- by **hot expression** from the dried, ripe seeds of *Linum usitatissimum* (Linaceae)
- yellow liquid when fresh but when exposed to air it gradually thickens and darkens in color and **form a hard varnish**



Fatty Acid Composition

- **36-50% α -linolenic acid (18:3 ($\Delta^{9,12,15}$))**
- 23-24% linoleic acid (18:2)
- 10-18% oleic acid (18:1)
- 7% stearic acid (18:0)
- 6% palmitic acid (16:0)



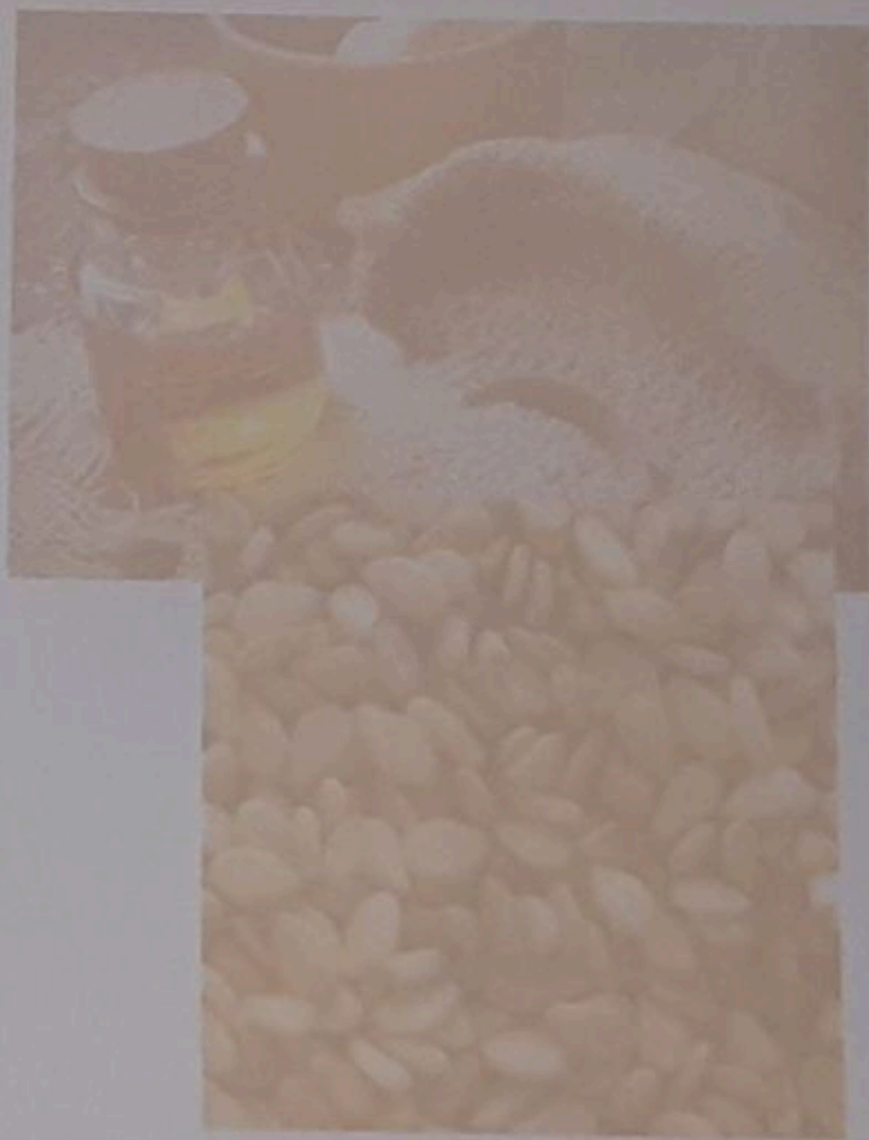
Uses

- Linseed oil is not suitable for food because of its **disagreeable taste**
- Because of its' high content of unsaturated fatty acid composition
 - it is a **drying oil** and thus important in paint and varnish production
- Linseed cake is valuable **cattle food**

- After hydrolysis it is very effective antibacterial agent and is used for wound dressing
 - active against *staphylococcus aureus* strains resistant to antibiotics
- reduces both the serum cholesterol level and blood pressure
- ω -3 FAs inhibit formation of thromboxane A_2 , which is required for platelet aggregation and clot formation
- exerts anti-inflammatory activity so help in rheumatoid arthritis and gastrointestinal inflammation

Sesame Oil

- The seeds of *Sesamum indicum* (Pedaliaceae) by expression or extraction
 - Fatty acid composition
 - 7-12% palmitic acid (16:0)
 - 3-6% stearic acid (18:0)
 - 35-50% oleic acid (18:1)
 - 35-50% linoleic acid (18:2)
 - The fixed oil also contains a considerable mounts of plant sterols, tocopherols (vitamin E).

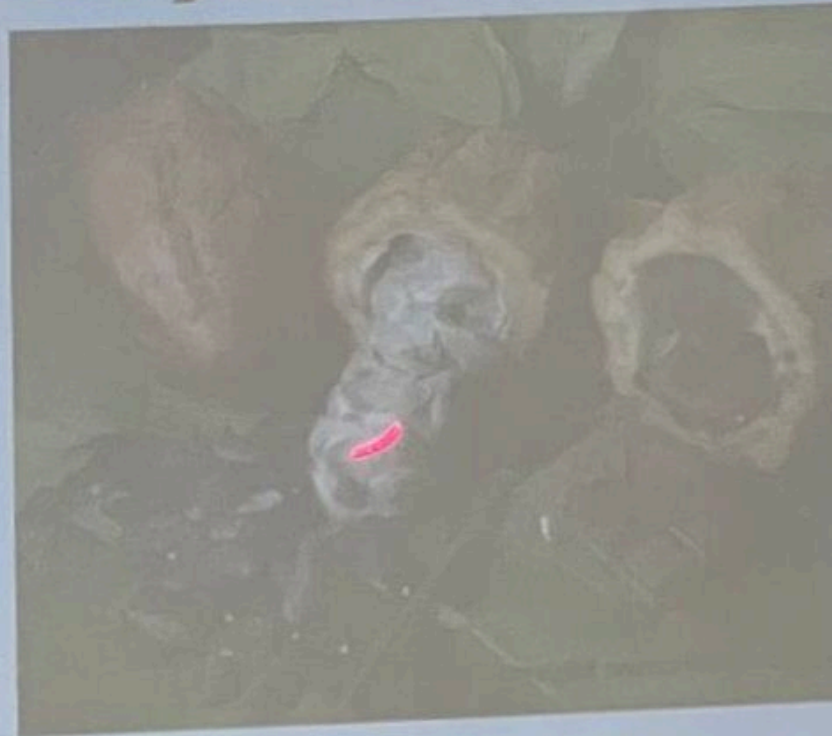


- Uses:

- As mild laxative
- In cosmetics
- As vehicle for drugs
- Sesame oil is commonly used to adulterate other oils

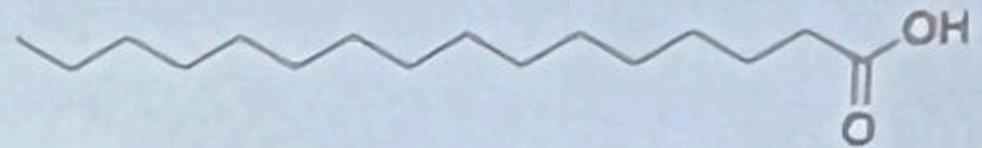
Theobroma Oil (Cocoa Butter)

- obtained by **hot expression** from the roasted kernels of ***Theobroma cacao*** (Sterculiaceae)
- yellowish white solid
- has a relatively sharp melting point: **30-35°C**



Fatty Acid Composition

- **25-29% palmitic acid (16:0)**
- **32-37% stearic acid (18:0)**
- **34-36% oleic acid (18:1)**
- 2% linoleic acid (18:2)

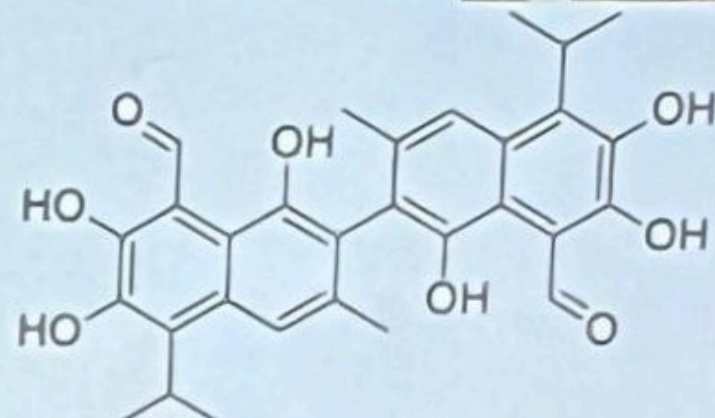


Uses

- For the preparation of **suppositories** and **lipsticks**
- Most expensive of the commercial fixed oils
- May be adulterated with waxes, stearin, animal and vegetable fats

Cotton Seed Oil

- Hot expression from the seeds of various *Gossypium* (Malvaceae).
- High temperature is employed in the extraction of cotton seed oil, in order to inactivate **gossypol**, which can cause **sterility in males**
- Gossypol is used as male contraceptive in China
- Cotton seed is a pale-yellow, semi-drying oily liquid



Fatty Acid Composition

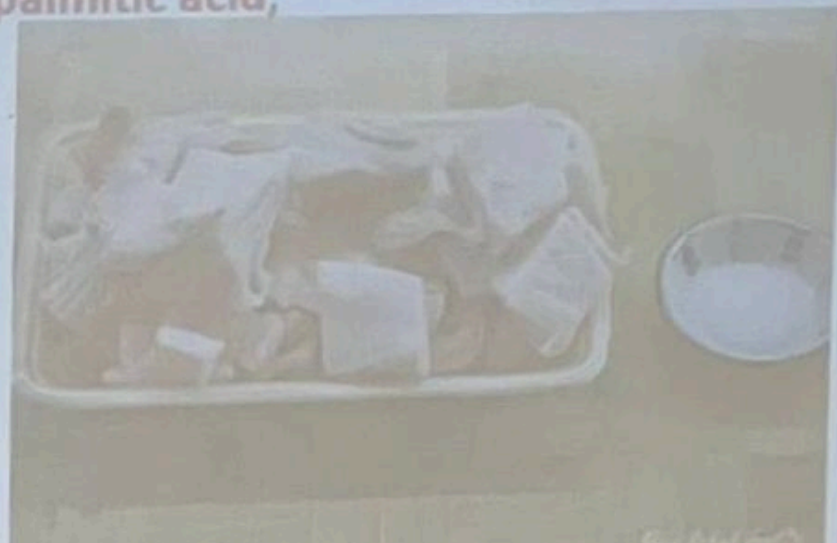
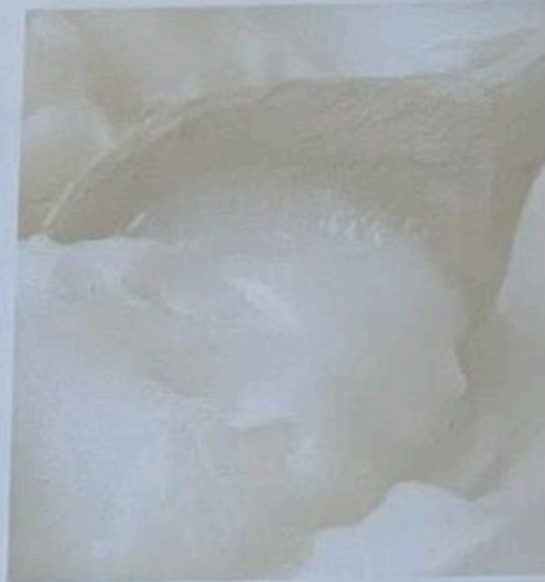
- 1-3% myristic acid (14:0)
- **18-45% palmitic acid (16:0)**
- 1-9% stearic acid (18:0)
- 10-32% oleic acid (18:1)
- **31-53% linoleic acid (18:2)**

Use:

- As vehicle for injections
- After hydrogenation, as substitute for lard.

Lard

- Is the purified internal fat of the hog (*Sus scrofa*: Suidae)
- is a soft, white fat with nonrancid taste
- Its melting point is from 34-41°C
- The fatty acids occurring in lard are:
 - mainly (65%) saturated ones such as **myristic acid**, **palmitic acid**, **stearic acid**
 - about 35% unsaturated ones such as **oleic acid**
- can easily get rancid and benzoin is added as a preservative
- used as ointment base



Code Liver Oil



- prepared from the fresh liver of the cod (*Gadus collarias*: Gadidae)
 - The cod livers which contains about 50% oil, is removed immediately after the fish are caught
 - The livers are then transferred to a steamer where the oil will be liquefied by the high temperature and separates
 - The oil is rapidly heated to 77 °C
 - Various reagents are added
 - removes impurities and
 - cause the dissolution of liver tissue fragments
 - The resulting oil is centrifuged
 - The above process is repeated twice and the oil is water-washed



- Drying is done in vacuum drying towers
- Winterization to remove stearin
- Deodorization
 - is carried out by steaming the oil under vacuum to remove aldehydic and ketonic impurities, which otherwise would impart disagreeable odor
- Standardization
 - After blending, the **vitamin A** and **vitamin D** content of the oil is determined
 - In BP, for instance, 1 gram of cod liver oil is required to **have 600 Units of vitamin A** and **60 Units of vitamin D**

- it is not allowed to add any artificial vitamin A and vitamin D into cod liver oil. Vitamin E, however, could be added

- Cod liver oil is composed of glycerides of :

- **unsaturated fatty acids (85%)** and saturated fatty acids (15%)
- Besides the common fatty acids, ω -3 series fatty acids such as 20:5 ($\Delta^{5,8,11,14,17}$) **Eicopentanoic acid** and 22:6 ($\Delta^{4,7,10,13,16,19}$) **docosaheaxaenoic acid DHA**

- Uses:

- Know to increase cognitive function of the brain
- As source of vitamin A and vitamin D

WAXES

- Waxes are mixtures, simple esters of long chain **usually saturated fatty acids** ($C_{14}-C_{26}$) and long chain alcohols ($C_{18}-C_{36}$)
- Waxes are solid at room temperature but soften and melt with heat
- Waxes are abundant in plants leaves' cuticle layer.
- commercially they are obtained from very few sources
- Animal and insect waxes are also used

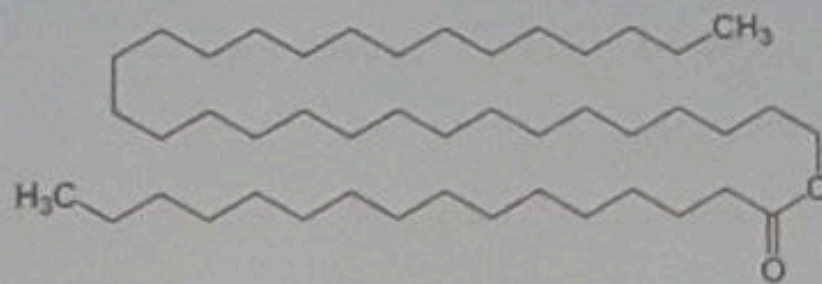
Beeswax (yellow wax)

- is the purified wax from the honey comb of bee (*Apis mellifera*: Apidae).
- Honeycomb, after separation from the honey, is melted in water, then cooled and remelted and finally strained and allowed to harden in mold



Honeycomb is composed of

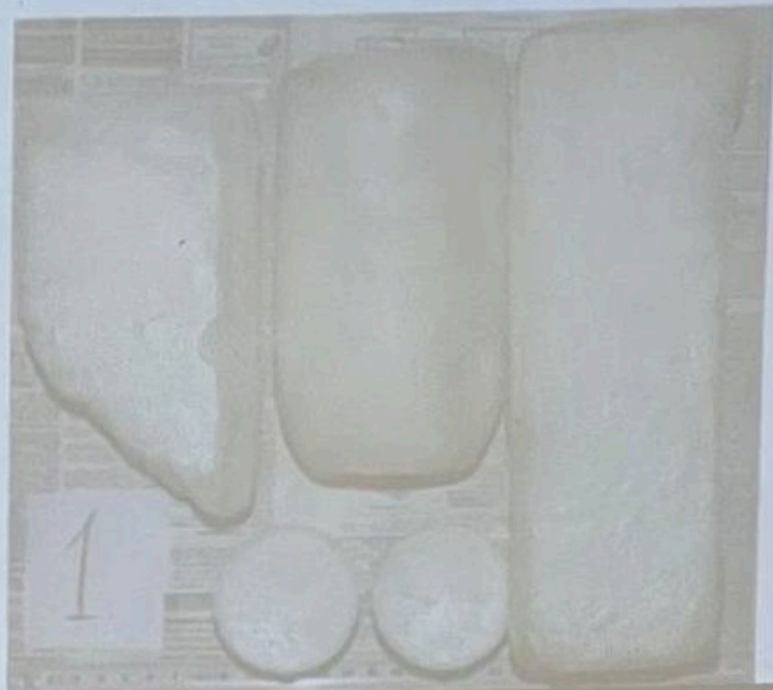
- myricyl palmitate $C_{15}H_{31}COOC_{30}H_{61}$
- myricyl cerotate $C_{25}H_{51}COOC_{30}H_{61}$
- similar high molecular weight esters



- yellow wax is used as
 - stiffening agent
 - base in plasters
 - in the manufacture of candles, cosmetics and polishes.
- White wax (bleached beeswax)
 - White wax is **bleached yellow wax**.
 - Bleaching is carried out by repeated exposure to sunlight
 - White wax is employed in ointments and cold creams

Spermaceti

- is a waxy substance obtained from the head of the sperm whale (*Physeter macrocephalus*: Physeteridae)



- valued for its quality emollient property and commonly added in **cold creams**
- sperm whale has become an endangered species
 - **natural spermaceti is no more available**
- Instead, synthetic spermaceti (cetyl esters wax) is used
 - Synthetic spermaceti is a mixture consisting primarily of esters of saturated fatty alcohols (C_{14} to C_{16}) and saturated fatty acids (C_{14} to C_{18})

Carnauba Wax

- Obtained from the leaves of *Copernicia prunifera* (Arecaceae).
- It contains chiefly **myricyl cerotate**
- Appreciated for its hardness and high melting point
- commonly used in the manufacture of candles and polishes and as beeswax substitute

